

Supplementary Material: CL-TAV Contracts and Core Mechanisms

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Abstract— This supplement records the accepted core mechanisms and the S0–S10 duty map for the TAES first draft. It does not change ALG-CLTAV-01 semantics and does not report experiments.

I. Scope

Paper Algorithm 1 is ALG-CLTAV-01. Paper-facing Algorithms 2–4 below are ALG-CLTAV-05, ALG-CLTAV-06 and ALG-CLTAV-07. Wrappers ALG-CLTAV-02/03/04 and typed records remain in the repository algorithm package; they are not a second contribution. Chinese S0–S10 glosses stay in the repository wrapper and are not copied here.

II. S0–S10 duty map

Step	Interface	Duty / exception
S0	—	initialize Γ, η, H ; distinct k , charges, retry
S1	IF-RESOURCE-STOP	already decided Admit or P-stop
S2	IF-PRED-OBS	TEST-scoped prediction gap
S3	IF-SELECT-ADMIT	exclusive kind; freeze snapshot
S4	—	charge once
S5	IF-EXECUTE-RECORD	effect class
S6	IF-OBS-INTERPRET	timing kernel ALG-CLTAV-06
S7–S8	—	classify effect; sole KNOWN writer
S9	IF-HIST-UPDATE	snapshot + postSummary
S10	IF-RESOURCE-STOP, IF-EQUIV	P-stop or continue to S1

III. Core mechanisms

The following modules are included by path from the accepted algorithm directory. Display captions keep the stable ALG-IDs.

Algorithm 1: SELECTANDADMIT: test selection and resource admission (ALG-CLTAV-05, S3)

Require: Γ ; η ; current H ; action library L ; typed $pred$
(PredictionResult)

Ensure: a Decision: kind ACTION / EXIT / SPEC-ERROR
with actionKind, actionId, classesUsed and reason

// state eligibility is independent of
affordability; under UNKNOWN only an
eligible Recover qualifies, never a Prep

- 1 $eligible(t) \leftarrow (\Gamma.qStatus = UNKNOWN ? (t.kind = RECOVER \wedge t.eligibleUnknown) : (t.kind \neq RECOVER \wedge t.enabledAt = \Gamma.currentSummary));$
- 2 $A \leftarrow \{t \in L : eligible(t)\};$
// affordability is decided by
IF-SELECT-ADMIT, not assumed from
eligibility
// a global or per-TEST projection gap
withholds TEST scoring and keeps the gap
identity; it does not itself block
eligible Prep/Recover
- 3 $testGap \leftarrow \text{false};$
- 4 **if** $pred.status = GAP$ **then**
- 5 $C \leftarrow \{\}; testGap \leftarrow \text{true};$
- 6 **else**
- 7 $C \leftarrow \{C_t\}$ with
 $C_t \leftarrow \{H \cap C : C \in pred.classesByTest[t], H \cap C \neq \emptyset\}$
 for each TEST $t \in A$;
- 8 **if** a TEST $t \in A$ has declared classes but $C_t = \emptyset$ **then**
- 9 $C \leftarrow \{\};$
- 10 $testGap \leftarrow \text{true};$
- // IF-SELECT-ADMIT is the single admission
 authority: affordability and A1-A5;
 exclusive kind; one-step minimax
 $s(t) = \max_{C \in C_t} |C|$ over currently valid
 nonempty classes of affordable
 distinguishing TESTs $D \cap F$ only; cost/id
 tie-break; never $\arg \min \emptyset$; Prep/Recover
 are not observation minimax;
 uninformative TESTs are not A2
- 11 $(k, S, t^*, admit) \leftarrow$
 $IF-SELECT-ADMIT(A, \Gamma.currentSummary, C, \Gamma, \eta, H);$
- 12 **if** $k = ADMIT$ **then**
- 13 **if** $testGap$ and $admit = A3$ **then return**
 Decision(kind = SPEC-ERROR, reason = $pred.reason$
 or per-test prediction gap);
- 14 **return** Decision(kind = EXIT, reason =
 the A2-A5 admission reason);
- 15 **else if** $k = PREP$ **then**
- 16 **return** Decision(kind = ACTION, actionKind =
 PREP, actionId = t^* , classesUsed = S);
- 17 **else if** $k = RECOVER$ **then**
- 18 **return** Decision(kind = ACTION, actionKind =
 RECOVER, actionId = t^* , $\cdot$);
- 19 **else if** $testGap$ **then**
- 20 **return** Decision(kind = SPEC-ERROR, reason =
 $pred.reason$);
- 21 **return** Decision(kind = ACTION, actionKind =
 TEST, actionId = t^* , classesUsed = C_{t^*});

Algorithm 2: INTERPRETTIMEDOBSERVATION:
timed observation kernel (ALG-CLTAV-06, S6
kernel)

Require: the single ownership result and the raw measurement interval from IF-OBS-INTERPRET; SelectSnapshot ξ ; named obligation inputs N_r (allowed interval) and D_r (physical domain); declared match policy Match_r ; declared uncertainty ε

Ensure: a TimingResult: verdict in {PASS, FAIL, INCONCLUSIVE, ERROR}; the single pairing evidence; the interval used; validity

// resolve N_r , D_r and ε from the named obligation and instance inputs, not from free symbols

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1 pair  $\leftarrow \text{Match}_r(\text{ownership}, \xi)$ ;
  // one pairing result feeds T2 and the
  // no-response rule
2 if pair is ambiguous then return
  TimingResult(verdict = ERROR);
3 if pair is a legitimate trigger/response pair then
4    $M \leftarrow \text{measurementInterval} \cap D_r$ ;
5   if  $M = \emptyset$  then return
     TimingResult(verdict = ERROR);
6   if  $M \subseteq N_r$  then
7     return TimingResult(verdict = PASS, evidence =
       pair,  $M$ );
8   else if  $M \cap N_r = \emptyset$  then
9     return TimingResult(verdict = FAIL, evidence =
       pair,  $M$ );
10  return TimingResult(verdict =
    INCONCLUSIVE, evidence = pair,  $M$ );
    // open/closed endpoints preserved
// a missing response forms a no-response
// only after a still-active instance
// reaches its deadline; this round not
// seeing it is not a timeout
11 if the obligation is still active and expired then
    // use the earliest admissible elapsed
    // horizon (a scalar), with the
    // requirement's boundary convention
12 if  $N_r$  is closed at  $U_r$  then
13   expired  $\leftarrow \text{lowerHorizon} > U_r$ ;
14 else
15   expired  $\leftarrow \text{lowerHorizon} \geq U_r$ ;
16 if expired then return TimingResult(verdict = FAIL);
17 return TimingResult(verdict = INCONCLUSIVE);
18 return TimingResult(verdict = ERROR);
    // a missing pair is not a point observation

```

Algorithm 3: COMMITCOMPATIBLEUPDATE:
history-compatible update and summary commit
(ALG-CLTAV-07, S9)

Require: SelectSnapshot ξ ; normalized Outcome z from the
resolving module; Γ ; η ; H ; precondition enforced by
the caller ALG-CLTAV-01: the S9 guard holds (a valid
observation class or a confirmed successor summary)

Ensure: a tagged UpdateResult ($status, \eta', H', \Gamma'$) with status
OK or SPEC-ERROR; Γ' initialized from Γ

```

1  $\Gamma' \leftarrow \Gamma$ ;
  //  $\Gamma'$  is a copy of  $\Gamma$ ; no field is left
  // undefined
  // a stale or inconsistent history version
  // fails closed before any update
2 if  $\xi.historyVersion \neq \eta.version$  then return
  UpdateResult(status = SPEC-ERROR,  $\eta, H, \Gamma'$ , reason =
  history version mismatch);
3 if  $z.I_z$  is a valid observation class then
4    $(\eta_c, H_c) \leftarrow$ 
    IF-HIST-UPDATE( $\eta, H, \xi.actionId, \xi.qUsedAtSelect, \xi.classesUsedAtSelect, \xi.historyVersion, z.I_z, z.postSummary$ );

    // IF-HIST-UPDATE is the sole history
    // writer; it must agree with the
    // intersection and the snapshot version
5   if  $H_c \neq H \cap z.I_z$  then return UpdateResult(status =
    SPEC-ERROR,  $\eta, H, \Gamma'$ , reason =
    intersection mismatch);
6    $\eta' \leftarrow \eta_c; H' \leftarrow H_c$ ;
7 else
8    $(\eta', H') \leftarrow$ 
    IF-HIST-UPDATE( $\eta, H, \xi.actionId, \xi.qUsedAtSelect, \xi.classesUsedAtSelect, \xi.historyVersion, NONE, z.postSummary$ );
     $H' \leftarrow H$ ;
    // a confirmed successor summary with no
    // valid observation advances history but
    // preserves  $H$ 
9 if  $\Gamma'.qStatus$  is KNOWN and  $z.summaryConfirmed$  is true
  and  $z.postSummary$  is confirmed then
10   $\Gamma'.currentSummary \leftarrow z.postSummary$ ;
    // this is the sole commit of the
    // confirmed successor summary
11 return UpdateResult(status = OK,  $\eta', H', \Gamma'$ );

```

IV. AI and third-party boundary

Drafting assistance was used for English wording and LaTeX layout. No standard figure is reprinted. No CRS row listing is published as supplementary data. Open-artefact licence and repository DOI remain AUTH-OPEN.